Waste Management 60 (2017) 140–150

Contents lists available at ScienceDirect

Waste Management

journal homepage: www.elsevier.com/locate/wasman

Process development for cigarette butts recycling into cellulose pulp

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ARTICLE INFO

Article history:
Received 2 May 2016
Revised 6 September 2016
Accepted 10 October 2016
Available online 5 November 2016

Keywords:
Cigarette butts
Solid waste
Cellulose acetate
Recycling
Cellulose pulp

ABSTRACT

Cigarette butts, which are usually thrown on the ground or into ordinary bins, have been recognized as toxic residues since may contain cigarette contaminants and chemicals produced during combustion. Therefore, contaminants in cigarette butts can be leached by rain into surface water and thereby contaminate the environment. In Brazil, according to the National Policy on Solid Waste, all residues must be disposed of in an environmentally friendly manner. Although cigarette butts are not mentioned in the law, due to their characteristics, they may be classified as hazardous waste. At the University of Brasilia, a cellulose pulp production process from cigarette butts has been developed employing alkaline pulping. This process is presented as an alternative to environmentally friendly final disposal of this residue. During the process, a dark liquor is generated, which was found to contain lignin, carbonyls, metals, nicotine and specific tobacco nitrosamines. The dark liquor was treated by acidification to promote lignin precipitation, coagulation with chitosan and $Al_2(SO_4)_3$ to remove metals and organic compounds and ozonized to oxidize resistant chemicals. The dark liquor presented a high chemical oxygen demand (COD; 29,986 mg/L), which was partially removed by precipitation (20%), chitosan coagulation (66%) and ozonation (45.8%). As the remaining COD was still high, we proposed reusing the clarified effluent in alkaline pulping, which seemed to be the easiest and most efficient procedure with the lowest cost.

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1. Introduction

It is known that there are about 5000 constituents in cigarette smoke, which are distilled from tobacco or generated during burning. Among these chemicals, at least 150 of which 44 are present in large quantities in smoke, are recognized as being highly toxic, mainly because of their carcinogenic and mutagenic potential (Hoffmann et al., 1990). To diminish the absorption of these chemicals in the body, as well as their emission into the air, modern cigarettes have cellulose acetate filters, which allow for safer use by reducing or even eliminating those compounds from the smoke. Indeed, these chemicals are partially or fully retained by the cellulose acetate filter, allowing for somewhat safer cigarette consumption.

However, the consumption of cigarettes containing cellulose acetate filters generates tons of toxic cigarette butts annually, which are usually discarded on the streets of cities and are responsible for delivering toxic compounds into soils and limnic water. Indeed, the cellulose acetate filters of cigarette butts are saturated with toxic compounds, unburned tobacco, ash and paper, and some researchers have highlighted them as environmental pollutants because of inadequate disposal, as butts are typically discarded into ordinary trash and on the ground (Barnes, 2011; Moerman and Potts, 2011; Novotny et al., 2009; Slaughter et al., 2011). Thus, these compounds can contaminate soil and be leached by rain water and brought to surface water and contaminate aquatic environments.

These wastes are small, but are generated in large quantities. For instance, according to the *Anuário Brasileiro do Tabaco*, the consumption of cigarettes in Brazil was about 89.1 billion in 2012 (Poll, 2013). Considering the cigarette butt average weight of 0.4 g, about 35,500 t of this residue is generated annually in Brazil. In the same year, worldwide consumption of cigarettes was about 5.8 trillion, of which about 2.6 trillion were consumed in China, representing about 2,320,000 t of cigarette butts discarded in the environment.

In Brazil, the National Policy on Solid Waste was instituted by Law 12305/2010 (Brasil, 2010), which regulates solid waste disposal and segregation according to the composition of the residue. According to Law 12305/2010, cigarette butts can be classified as municipal and hazardous waste and their segregation and disposal

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http://dx.doi.org/10.1016/j.wasman.2016.10.013
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must be performed properly in order to prevent soil pollution in garbage dump and consequent pollution of surface and underground water. For this reason, in urban areas, cigarette butts have begun to be collected in special containers. This responsibility may not only be charged upon the consumer or governments. Indeed, some authors underline that "under the environmental principle of Extended Producer Responsibility, tobacco product manufacturers may be held responsible for collection, transport, processing and safe disposal of tobacco product waste (TPW)" (Curtis et al., 2016). However, the final destination of cigarette butts is still a challenge in this field because, until the present work, there has not been a recycling technology available, meaning that cigarette butts are usually incinerated or disposed of in garbage dump.

In order to promote the environmentally friendly disposal of cigarette butts, our research group has developed a technology to recycle them for cellulose pulp production (PI 0305004-1 A) (Costa et al., 2005). The technology consists of separating all the cellulose content in the cigarette butt using four different processes: (i) dissolving the ash and all chemicals adsorbed in the filter; (ii) hydrolyzing cellulose acetate into cellulose; (iii) removing lignin from the remaining tobacco and liberating its cellulose; and (iv) disaggregation of the remaining paper. After filtering the resulting mixture, we obtained a cellulose pulp, which can be used in the paper industry, and a viscous dark liquor, which requires treatment. In the present work, we describe this method of cigarette butt processing, the characterization of the cellulose pulp, the analysis of the dark liquor and its possible treatments.

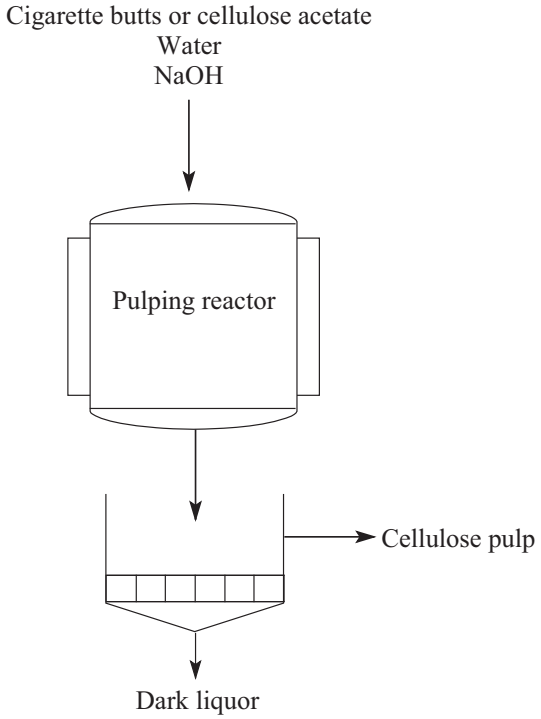


Fig. 1. Pulp production flowchart.

2. Materials and methods

2.1. Cellulose pulp production

Cigarette butts were collected from cigarette butt collectors located in restaurants, bars and buildings from the Campus of the University of Brasilia and surroundings. The starting mixture contained cigarette butts from different blends and with different amounts of remaining tobacco and paper. Wastes from filter production in the cigarette industry were also used, which contained only cellulose acetate and paper. For the reaction, NaOH was used at a 1% concentration in a water solution to catalyze the reaction. The pulping or cooking process for each fiber (cigarette butts and cellulose acetate) occurred in a period of 3 h, at boiling temperature, in a pilot-scale stainless steel reactor with a 50 L capacity. The reactor was heated with a flame of liquefied petroleum gas (LPG) and the temperature was monitored by a digital thermocouple. A flowchart of the overall process is shown in Fig. 1.

Samples of pulps obtained from cigarette butts and cigarette filter wastes were sent to the Instituto de Pesquisas Tecnológicas (IPT) in São Paulo, Brazil, to evaluate the physical and optical parameters of the cellulose for commercial use as paper. The samples were refined in a Jokro mill for 25 min and the extent of refining was determined by the Schopper-Riegler degree (°SR). In addition, sheets from these samples were processed in TAPPI (Technical Association of Pulp and Paper Industry) forming equipment in order to evaluate the weight, thickness, brightness, tensile strength and elongation. In Table 1 are listed the standard methods used.

Table 1
Methods of analysis used on the cellulose pulp.

Analysis	Standard method
Cellulose pulp	
Refining by a Jokro refiner	NBR[a] 14346:1999
Sheet formation	NBR 14479:2000
Paper sheets	
Grammage	NBR 14527:2000
Thickness	NBR 14527:2000
Whiteness	NBR NM ISO 2470:2001
Tensile index	NBR 14527:2000
Stretching	NBR 14527:2000

[a] NBR: standard of the Brazilian Association of Technical Standards.

2.2. Dark liquor characterization and treatment

Wastewater from cellulose pulp production (dark liquor) using cigarette butts was characterized by measuring the organic matter by determining the chemical oxygen demand (COD), biochemical oxygen demand (BOD) and total organic carbon (TOC). In addition, the settleable solids, the presence of lignin and the content of metal ions and some organic compounds usually found in cigarette smoke and tobacco, such as carbonylated molecules, tobacco specific nitrosamines and nicotine were assessed.

COD determination was carried out by closed reflux colorimetry using a photocolorimeter from Alfakit (model AT10P), in accordance with Standard Methods for the Examination of Water and Wastewater (SMWW), method SMWW 5220D (American Public Health Association, APHA, 1999). BOD determination was carried out by a respirometric method for 5 days at 20 °C (Oxidirect Aqualytic, model Al606), in accordance with SMWW 5210D (APHA, 1999). TOC was determined by catalytic oxidation at 680 °C in a carbon analyzer from Shimadzu (model LCPH/CPN PC), according to SMWW 5310B (APHA, 1999). The samples had been previously filtered through a microfilter Millipore Millex-HV with a porosity of 0.45 μm. Settleable solids were determined by the standard method ABNT/NBR 10561/1988 (ABNT, 1988).

The precipitation of lignin from the dark liquor was performed by acidification, as suggested by several studies in the literature by Minu et al. (2012), Mussatto et al. (2007) and García et al. (2009). As the acidifying agents, HCl (6 N), H_3PO_4 (concentrated and diluted 2% v/v), CH_3COOH (concentrated) and H_2SO_4 (diluted 2% v/v) were used. The acidification agent was added to the dark

liquor in different amounts in order to achieve pH values of 1–9, and the precipitation efficiency was determined by comparing the COD of diluted supernatants and the starting value, yielding the percentage removal of organic matter. The lignin removal from the starting dark liquor was confirmed by spectroscopic studies to investigate the characteristic lignin bands by infrared spectroscopy using an attenuated total reflectance (ATR) cell in FT-IR apparatus (Shimadzu, model IR Prestige-21).

The precipitation supernatant was referred to as the clarified effluent and, after determining the best pH level for precipitation, coagulation tests were carried out using $Al_2(SO_4)_3$ and chitosan as the coagulant agents. The assays were adapted from tests performed by Vaz et al. (2010), Ahmad et al. (2011) and Guimarães (2013). In an Erlenmeyer flask (125 mL capacity), the clarified effluent (75 mL) was mixed with the necessary amount of coagulant to promote coagulation. It is important to note that care was taken not to exceed 7.5 mL of coagulant, equivalent to 10% of the clarified effluent volume. The mixture was shaken using an orbital shaker table (Tecnal, model TE-141). Coagulation was performed by rapid mixing (coagulation) at 180 rpm for 2 min and slow mixing (flocculation) at 60 rpm for 15 min. Then, the mixture was allowed to rest (decantation) for 20 min. Assays were carried out at different pH values ranging from 6 to 9 and different concentrations of coagulants (presented in Table 2) were used according to the tests performed by Guimarães (2013). The coagulation efficiency was assessed by COD analysis.

We also evaluated the possibility of mineralizing the organic matter by ozonation. Indeed, ozonation was carried out as a pre-oxidation treatment (directly in the dark liquor) and as an effluent post-oxidation process (in the precipitation and coagulation supernatants). The ozonation tests were performed using an installation adapted from Guimarães et al. (2010) and Houshyar et al. (2012), depicted in Fig. 2. An air compressor (Millipore, model WP6111560) fed a mini ozone generator (Di Acqua, model Di Acqua $FlexO_3/ClO_2$). The gas flow from the ozone generator was adjusted to 0.5 L/min and its ozone content (mass flow) was determined as 1 mg/min by titration, as described by standard method SMWW 2350D (APHA, 1999). The ozone/air flow from the ozone generator was forced into a sequence of three traps, as described in Fig. 2. In the first trap, the ozone was bubbled through a bubble diffuser into the effluent (150 mL) under constant magnetic stirring. The exhaust from the first trap was directed into the second trap, which was empty, in order to collect any liquid or foam swept along by the current. Finally, the gas was bubbled into a 2% (v/v) KI solution (200 mL) in the third trap. After titration of this KI solution, according to standard method SMWW 2350D, it was possible to quantify the unreacted ozone, and, thus, by mass balance, approximately determine the amount of ozone that reacted with the effluent. The ozonation tests took place over a period of 4 h and samples were collected every hour to measure the pH and temperature and perform COD analysis. This procedure was carried out using the following effluents: (1) the dark liquor, (2) clarified effluent and (3) supernatant from the best coagulation assay. For each sample, tests were conducted at different starting pH levels: 9, 10, 11 and 12. For samples (2) and (3), it was necessary to add 0.5 mL of Aldrich Antifoam A5633 defoamer solution (1 mL of antifoam dissolved in 30 mL of propylene glycol).

To investigate the presence of carbonylated compounds, the dark liquor was analyzed by high performance liquid chromatography (HPLC), adapting the methodology proposed by British American Tobacco (BAT) (BAT, 2008a) to assess carbonylated molecules in cigarette smoke. For this analysis, dark liquor (1 mL) was diluted in distilled water to a final volume of 100 mL. Then, 1 mL of this solution that was mixed with 1 mL of 2,4-dinitrophenylhydrazine (DNPH) solution in methanol (2.5 g/L) and 1 mL of buffer pH 4 solution (10 M NaOH/glacial acetic acid). This mixture was homogenized and then filtered through a Millipore Millex-HV microfilter with a porosity of 0.45 μm and injected into chromatograph (Shimadzu, model CTO-20A UV–vis), with a detector at a wavelength of 380 nm, equipped with Shim-Pack VP-ODS (C-18, 250 mm, 4.6 mm inner diameter). The flow rate was 1.5 mL/min, the injection volume was 20 μL, and the column oven temperature was 30 °C. The total run time was 35 min (see Table 3) using a gradient of two mobile phases: (A) 59% water (vol.%), 30% acetonitrile, 10% tetrahydrofuran and 1% isopropanol and (B) 33% water (vol.%) 65% acetonitrile, 1% tetrahydrofuran and 1% isopropanol. The carbonylated compounds investigated were formaldehyde, acetaldehyde, acetone, propionaldehyde, methyl ethyl ketone, crotonaldehyde and butyraldehyde. It was prepared a "standard solution" adding 0.01 mL of each of these carbonylated compounds and completing to 100 mL with a solution of DNPH in methanol (2.5 g L^{-1}). To check the retention time of each carbonylated compound, different

Table 2
Coagulant concentrations used in the coagulation experiments.

Coagulant	Concentration (g/L)
$Al_2(SO_4)_3$ (aqueous solution)	0.10, 0.15, 0.20, 0.25 and 0.30
Chitosan (HCl 0.5% aqueous solution)	0.05, 0.08, 0.20, 0.50 and 0.80

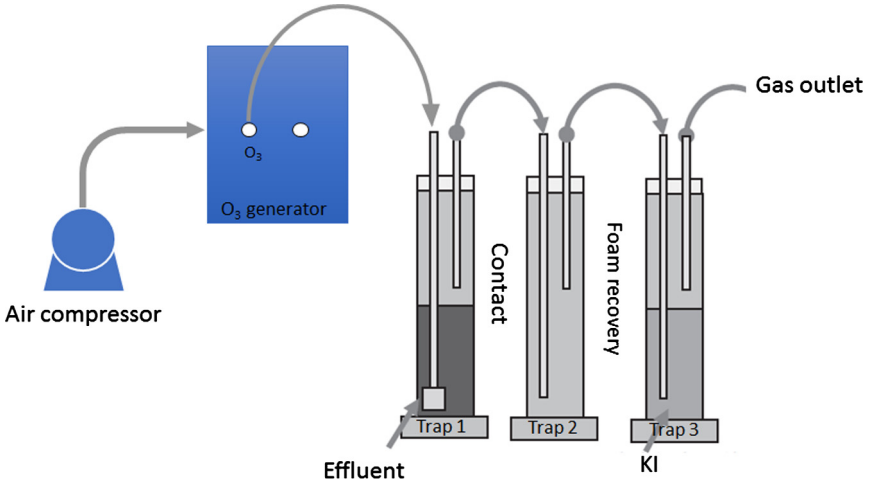


Fig. 2. Ozonation apparatus (adapted from Guimarães et al. (2010)).

Table 3
Mobile phases used during HPLC analysis to investigate carbonylated compounds.

Time (min)	Mobile phase A (%)	Mobile phase B (%)
0	100	0
8	70	30
20	47	53
25	15	85
27	100	0
35	100	0

Mobile phases: (A) 59% water (vol.%), 30% acetonitrile, 10% tetrahydrofuran and 1% isopropanol and (B) 33% water (vol.%) 65% acetonitrile, 1% tetrahydrofuran and 1% isopropanol.

runs were made using the same analysis conditions as proposed by BAT (2008a) standard method: (1) pure methanol; (2) the standard solution of carbonylated compounds, (3) a mixture of standard solution of carbonylated compounds (1 mL) and 1 mL of one of the carbonylated compound; and (4) a mixture of 1 mL DNPH solution, 1 mL of the carbonylated compound and 1 mL of the buffer solution.

We also investigated the presence of the elements arsenic, cadmium, lead, chromium, nickel and selenium in the dark liquor and the effluents from the different processes. The samples were previously digested using microwave energy and the EPA 3015A acid digestion method (EPA, 2007). Then, the samples were analyzed by inductively coupled plasma optical emission spectrometry, ICP-OES (Thermo Scientific apparatus, model ICAP 6000 Series).

The determination of tobacco-specific nitrosamines and nicotine in the dark liquor and the effluents from the different treatments was performed by high-performance liquid chromatography coupled to mass spectrometry (LC-MS) with a triple quadrupole and electrospray ionization, with Persuit XRS C18 150 × 2.0 mm column (Varian apparatus, model LC Series 212-LC and MS Series 310-MS). For the determination of tobacco-specific nitrosamines (TSNs), the methodology was the same as that proposed by BAT to quantify nitrosamines in mainstream smoke (BAT, 2008b). The studied nitrosamines were 4-(methylnitrosoamino)-1-(3-pyridinyl)-1-butanone (NNK) and N′-nitrosonornicotine (NNN). Standard solutions were prepared in HPLC grade methanol, which allowed dilution to construct calibration curves to quantify these TSNs in the dark liquor, clarified effluent, best coagulation supernatant and samples obtained from ozonation. For nicotine analysis, the method proposed by Franco (2014) was adapted. The mobile phase was 45% acetonitrile, 40% MilliQ water and 15% 5 mM phosphate monobasic buffer solution with the pH adjusted to 6.80. The flow rate was 1.0 mL/min, the injection volume was 20 µL, and the run time was 10 min. A calibration curve was constructed with nicotine solutions in Milli-Q water.

3. Results and discussion

3.1. Cellulose pulp production

Cigarette butts contains four different components: (i) a filter made from cellulose acetate with a myriad of chemicals adsorbed in it, (ii) unburned tobacco, (iii) ash and (iv) paper. In order to develop an environmentally friendly final disposal method for cigarette butts, a process was proposed to produce cellulose pulp, which can be used in the recycled paper market. Indeed, the main idea was to hydrolyze cellulose acetate into cellulose in the presence of strong alkali, depicted in Fig. 3. The use of a strong base may also remove lignin from unburned tobacco in the cigarette butt and thus liberate the cellulose from this residue, as well as dissolve the ash and disaggregate the remaining paper. Thus, at the end of the process, cellulose was obtained as a solid phase as well as a solution containing all the non-cellulose components of the cigarette butt.

Fig. 4 shows the FT-IR spectra obtained after cooking cigarette butts at 80 °C under different conditions: using distilled water for 6 h; NaOH solution (0.5%) for 10 min, 20 min and 150 min; and NaOH solution (1%) for 10 min. When cigarette butts were cooked using distilled water, no significant change in the FT-IR spectra was observed, even after 6 h (Fig. 4, curve 'a'). As shown in Fig. 4, it is possible to note a strong band at 1760 cm^{-1} and a shoulder at 1730 cm^{-1}, corresponding to C=O stretching from the acetate group, strongly suggesting the preservation of cellulose acetate. However, when the cigarette butts were cooked in NaOH solution (0.5%), even for short times (Fig. 4, curve 'b'), it was possible to observe a decrease in the band at 1760 cm^{-1}, as well as in the shoulder at 1730 cm^{-1}; these features completely disappeared

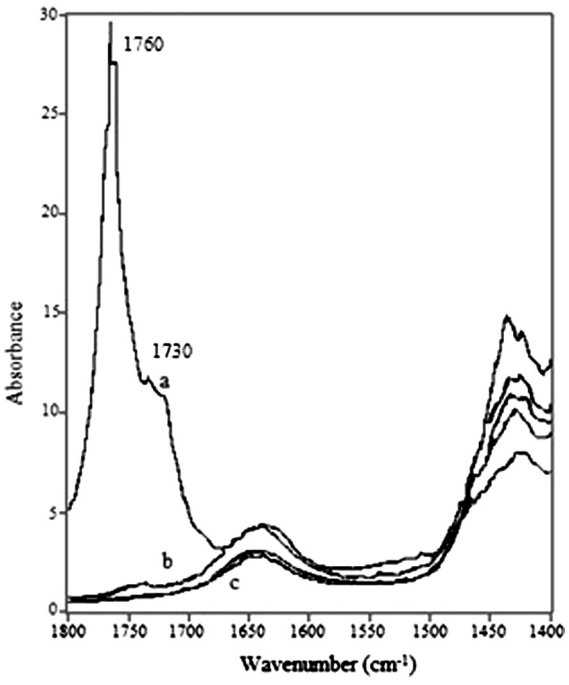


Fig. 4. FT-IR spectra of cigarette butts cooked at 80 °C under different cooking conditions: (a) distilled water for 6 h, (b) NaOH aqueous solution (0.5%) for 10 min, (c) NaOH aqueous solution (0.5%) for 20 min. Note that the spectra of cigarette butts cooked using NaOH aqueous solution (0.5%) for 150 min and NaOH aqueous solution (1%) for 10 min overlap with curve (c).

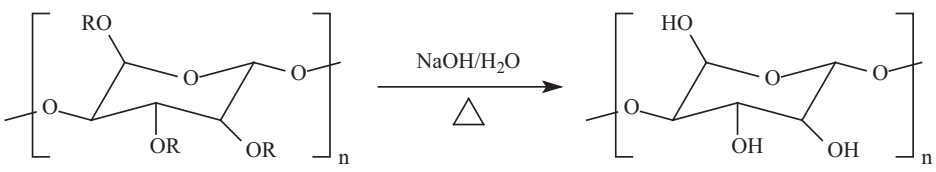


Fig. 3. Hydrolysis of cellulose acetate in alkaline medium (R = H or $COCH_3$).

with longer treatment times (Fig. 4, curve 'c'). It is worth mentioning that cooking cigarette butts in NaOH solution with a higher concentration (1%) led to the complete disappearance of these bands after 10 min (Fig. 4, curve 'c'). These results show that cellulose acetate is quickly hydrolyzed into cellulose in alkaline medium and that the reaction rate is dependent on the NaOH concentration.

In order to verify the potential use of this process to produce cellulose pulp, paper sheets were prepared using fibers disaggregated in a Jokro mill. Two different paper sheets were prepared: (a) using cellulose acetate with no previous treatment and (b) cigarette butts cooked in alkaline medium. Then, the main paper quality parameters were analyzed for both paper sheets; the results are presented in Table 4. As shown in Table 4, the degree of refining of cellulose pulp from cigarette butts and cellulose acetate was 46 °SR and 15 °SR, respectively, which indicate that, in previously cooked samples, greater entanglement of fibers occurs during paper formation and consequently better drainage take place, as previously shown by Campos et al. (2000). Moreover, as shown in Table 4, the paper sheet produced using cellulose acetate fibers with no treatment had very poor mechanical properties (traction index, tensile strength and stretching), which were significantly increased after alkaline cooking. These results suggest that the cellulose fibers become suitable for use as paper only after alkaline cooking. Thus, this indicates that the process proposed in this work is suitable for recycling cigarette butts into cellulose fiber that can be used in the paper industry, either pure or blended with cellulose pulps from various origins according to availability or the final desired properties. However, the low degree of whiteness indicates that the treatment was not sufficient to remove the residues deposited on the fibers during cigarette burning, thus, depending on the use of the fibers, a further bleaching process is required.

Table 4
Physical properties determined for papers prepared with cellulose acetate filters and cigarette butts.

Analysis	Samples	
	Cellulose acetate filter without pulping	Filters from cigarette butts pulped in alkaline medium
Schopper-Riegler grade (°SR)	14	46
Grammage (g/m^2)	67.2 ± 2.14	71.14 ± 1.98
Thickness (mm)	0.147 ± 0.004	0.173 ± 0.004
Whiteness (%)	84.8 ± 0.51	54.16 ± 0.20
Tensile strength (kN/m)	0.07 ± 0.01	0.80 ± 0.08
Traction index (kN m/kg)	1.09 ± 0.15	11.22 ± 1.83
Stretching (%)	0.67 ± 0.08	1.91 ± 0.23

Table 5
Dark liquor characteristics.

Parameter	Dark liquor	Effluent discharge conditions according to CONAMA 430/2011 (Brasil, 2011)
Temperature (°C)	33.1 ± 3.1	<40
pH	8.5 ± 0.4	5–9
Settleable solids (mL/L)	0.8 ± 0.04	⩽1.0
Floating materials	Absent	Absent
COD (mg/L)	29,986 ± 7037	Unspecified

3.2. Dark liquor characterization and treatment

It is worth mentioning that the collected cigarette butts were a very heterogeneous mixture. Indeed, it was possible to find cigarette butts from dozens of brands, each one with its specific original tobacco and blend of additives, and different amounts of remaining unburnt tobacco, as well as other possible contaminants, such as lipstick and saliva residue. Thus, it is reasonable to expect different chemical components in the dark liquor according to the starting mixture of cigarette butts. In order to study the dark liquor generated during the alkaline cooking of cigarette butts, 35 experiments were done with different starting materials; samples were taken after cooling and filtering the final mixture.

First, the main effluent parameters were determined and are presented in Table 5 as the average values obtained from the 35 dark liquor samples in comparison with the effluent release conditions stipulated by Brazilian law. As can be noted from the results presented in Table 5, the dark liquor had on average a very high organic content, even higher than those values usually observed in domestic sewage.

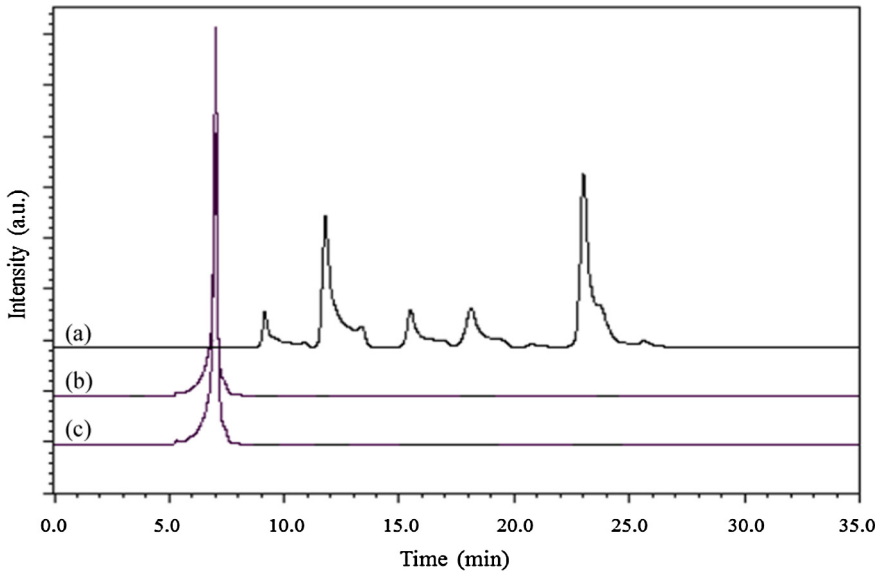


Fig. 5. HPLC Chromatograms: (a) Methanolic solution of standard aldehydes and ketones (formaldehyde, acetaldehyde, acetone, propionaldehyde, methyl ethyl ketone, crotonaldehyde and butyraldehyde) and DNPH; (b) dark liquor solution (1 mL 1:100 solution of dark liquor + 1 mL DNPH solution in methanol + 1 mL buffer pH 4); and (c) blank (1 mL DNPH solution in methanol + 1 mL buffer pH 4).

In sequence, different chemical analyses were carried out in order to understand the chemical composition of the organic content in the dark liquor. The scientific literature concerning cigarettes is usually focused in studying the chemical composition of cigarette smoke and, to the best of our knowledge, there are no references to the remaining chemicals in the filters and evidently nothing concerning the dark liquor obtained after cooking cigarette butts. According to BAT, it is possible to identify more than 5000 chemical components in cigarette smoke, as well as some metal ions (Hoffmann et al., 1990; Rodgman and Perfetti, 2009). However, most of these compounds are present in small amounts. The most important are classified according to their functional groups: aromatic and heterocyclic amines, carbonylated compounds, phenols, polycyclic aromatic hydrocarbons, carbon and nitrogen oxides, and ammonia. The complete list of these compounds is available in the literature (Hoffmann et al., 1990; Rodgman and Perfetti, 2009). Thus, we decided to investigate the presence of these compounds in the dark liquor using the analytical methods proposed by BAT.

Table 6
Contents of NNK, NNN and nicotine in the dark liquor in comparison to the literature data for cigarette smoke.

	NNK	NNN	Nicotine
Tobacco smoke (ng/cig[a]) (Hoffmann et al., 1990)	115.6	133.1	750,000
Dark liquor (ng/mL)	3057	1635	35,400,000

[a] cig: cigarette.

Table 7
Concentration of As, Cd, Cr, Ni, Pb and Se in the dark liquor determined by ICP-OES and the maximum values specified in Brazilian law for discharging effluents into limnic water.

Element	Content in dark liquor (mg/L)	Concentration limit allowed by Brazilian law in discharge effluents (mg/L) (Brasil, 2011)
As	0.04 ± 0.0003	0.5
Cd	ND	0.2
Cr	0.01 ± 0.0001	0.1 (Cr^{6+}) 1.0 (Cr^{3+})
Ni	ND	2.0
Pb	ND	0.5
Se	ND	0.3

ND: not detected.

The presence of aldehydes and other carbonylated compounds in the dark liquor was studied by liquid chromatography; the chromatograms are shown in Fig. 5. As shown in Fig. 5, no carbonylated compounds were detected in significant amounts when comparing the chromatogram obtained for dark liquor (Fig. 5b) to carbonylated compound standards (Fig. 5a; formaldehyde, acetaldehyde, acetone, propionaldehyde, methyl ethyl ketone, crotonaldehyde and butyraldehyde). The only peak observed in the dark liquor chromatogram was at a retention time of 6.992 min, also observed in the blank (Fig. 5c), which was related to the buffer required for analyzing the samples according to the method defined by BAT (2008a). The absence of these compounds usually detected in smoke is not surprisingly due to the high volatility and reactivity of aldehydes and ketones, such that they are not retained in the filter.

Tobacco-specific nitrosamines, such as 4-methylnitrosoamino-1-3-pyridinyl-1-butanone (NNK) and N′-nitrosonornicotine (NNN), as well as nicotine, were evaluated by liquid chromatography coupled with mass spectroscopy; the results are shown in Table 6. For comparison purposes, Table 6 also contains the values available in the literature for the content of these compounds in cigarette smoke (Hoffmann et al., 1990). As can be noted in Table 6, although there is no specification regarding these chemicals in the environmental law concerning wastewater, the contents of NNK, NNN and nicotine in the dark liquor were significant, especially taking in account the toxicity of these compounds. Comparing the relative content of these compounds in cigarette smoke and butts, it is possible to suggest that the filter is more effective at retaining nitrosamines than nicotine.

Finally, the presence of As, Cd, Pb, Ni and Se was studied by ICP/OES; the results are presented in Table 7. Table 7 also provides the maximum values allowed by Brazilian law for wastewater. According to Table 7, it was possible to detect only As and Cr in the dark liquor, probably because the other studied elements were at concentrations below the detection limit of the equipment. However, the As and Cr contents were at values below the maximum allowed under Brazilian law.

According to the values presented in Tables 5–7, under Brazilian law, the dark liquor produced from cigarette butt hydrolysis could

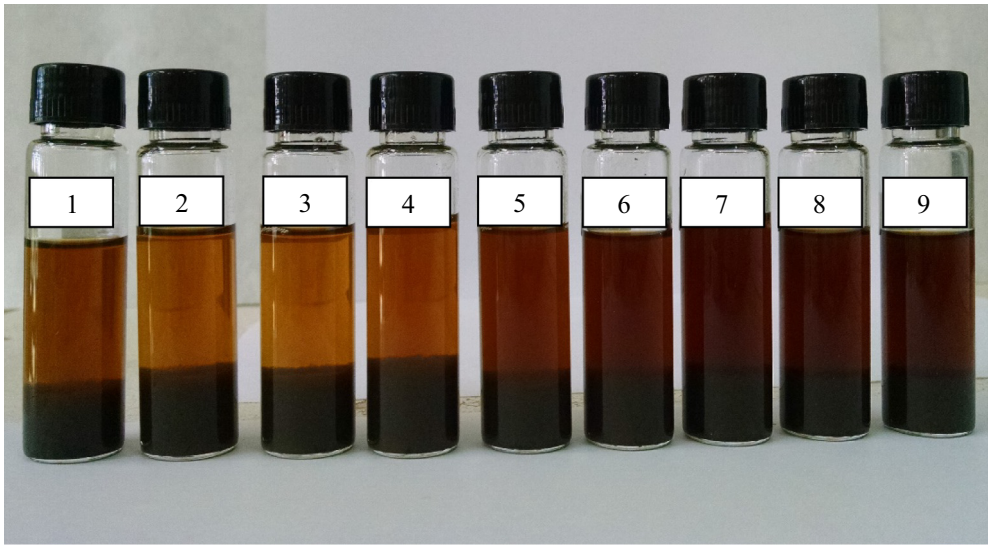


Fig. 6. Effect of pH on dark liquor color after acidification.

be discarded directly into a water body without further treatment. However, from an environmental awareness point of view, it would be unthinkable to disregard the high chemical oxygen demand (COD), especially taking in account the fact that highly toxic compounds were detected in the effluent. Thus, we decided to develop a clean solution for the cigarette butt recycling process.

Initially, it is important to highlight that COD measurement is the sum of biodegradable and resistant organic compounds. By analyzing biodegradable organic matter, the dark liquor showed an average biochemical oxygen demand (BOD) value of 9548 mg/L ± 2739 mg/L, indicating that most of the COD (29,986 mg/L ± 7037 mg/L) was due to resistant chemicals. According to Tchobanoglous et al. (2003), if the BOD/COD ratio for untreated wastewater is higher than 0.5, the effluent may be easily treatable by a biological processes. In our case, the BOD/COD ratio was 0.3, suggesting that treatment should be performed by physical-chemical processes.

In an initial attempt to develop a physical-chemical treatment for the dark liquor, an acid precipitation process was applied based on the methods of Minu et al. (2012), Mussatto et al. (2007) and García et al. (2009). Different organic and inorganic acids were used; the best precipitation results were obtained using HCl and H_2SO_4. As the H_2SO_4 market is highly regulated by Brazilian Federal Police, we decided to resume the study using HCl as the acidifying agent. Precipitation tests were performed at pH 1 to 9 to obtain higher COD removal. Fig. 6 shows the effect of pH on dark liquor color after acidifying the medium and Table 8 shows the COD removal in each pH. It is clear based on Fig. 6 that a significant decrease in color intensity occurred when acidifying the medium and that the best results were found when adjusting the pH in the range from 2 to 4. According to Table 8, the best result concerning a decrease in the COD value was achieved at pH 3, affording a removal of 20.1%. Similar behavior was observed by Minu et al. (2012), when using H_2SO_4 to clarify a black liquor obtained from processing rice straw. From now on, the supernatant obtained by this acidification process will be called the **clarified effluent**.

Table 8
COD results from dark liquor and the supernatants after precipitation at different pH levels.

Sample	COD removal (%)
Dark liquor	0.0
pH 1 supernatant	9.5
pH 2 supernatant	7.5
pH 3 supernatant	20.1
pH 4 supernatant	3.3
pH 5 supernatant	2.4
pH 6 supernatant	4.4
pH 7 supernatant	1.6
pH 8 supernatant	2.2
pH 9 supernatant	0.2

Aiming to understand what was removed after acidifying the dark liquor to obtain the clarified effluent, the non-volatile content from these effluents was isolated by evaporating the water in a Petri plate. Thus, after standing at room temperature and atmospheric pressure until a constant weight was achieved, the obtained films were dissolved in methanol and dropped on an ATR cell. After evaporating the alcohol, the film was analyzed by FT-IR. A similar procedure was used for the precipitate. Fig. 7 shows the FT-IR spectra of the films obtained from the dark liquor, the clarified effluent and the precipitate. According to Fig. 7, the spectra of the film obtained from the dark liquor (Fig. 7a) exhibits three strong bands at 1550 cm^{-1} and 1402 cm^{-1}, related to skeletal vibrations involving C—C stretching of mono-nuclear aromatic rings, and at 1340 cm^{-1}, characteristic of syringyl rings. It is also possible to observe a low intensity shoulder at 1260 cm^{-1} and a band at 1207 cm^{-1}, usually related to the guaiacyl ring. These three bands are usually attributed to lignin. Similar spectra were observed by Minu et al. (2012), studying a black liquor obtained from processing rice straw to obtain cellulose. A broad band around 3300 cm^{-1} was also observed in Fig. 7a, indicating the presence of alcoholic and phenolic OH groups, probably indicating the partial hydrolysis of lignin from tobacco during the cooking of cigarette butts. In the spectrum of the film from the supernatant (Fig. 7b), the phenolic OH groups were retained in this sample, which was not observed in the spectrum of the precipitate film (Fig. 7c). This result seems to indicate that only lignin was precipitated, confirming the data described in the literature regarding the removal of lignin by precipitation in acid medium (Minu et al., 2012), and that the partially hydrolyzed lignin remained in the supernatant.

Although the precipitation of lignin by acidification of the dark liquor removed up to 20% of COD, the average residual value of the clarified effluent was still very high, up to 23,959 mg/L, indicating other steps of treatment are still needed. Thus, the clarified effluent

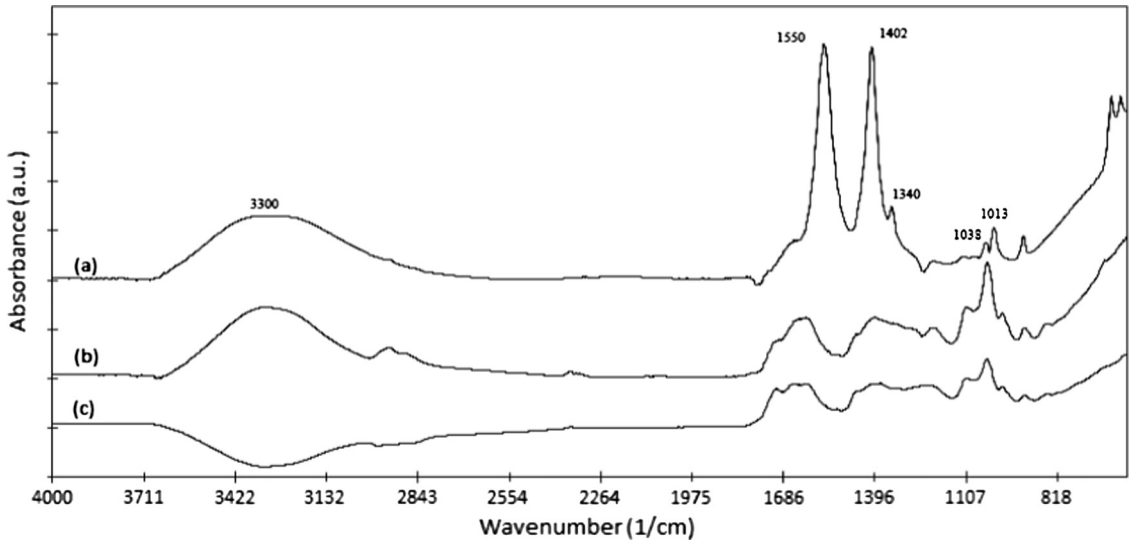


Fig. 7. ATR FT-IR spectra of films obtained from the dark liquor (a), the supernatant (b) and the precipitate (c) obtained after acidification to pH 3 with 6 N HCl.

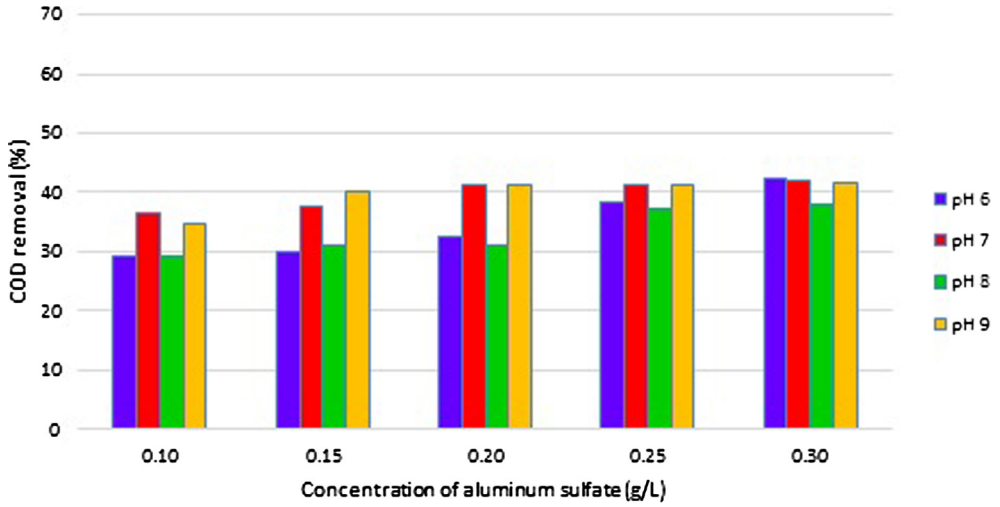


Fig. 8. COD removal by coagulation of the clarified effluent with $Al_2(SO_4)_3$.

was coagulated with $Al_2(SO_4)_3$, with the pH adjusted from 6 to 9 and the main results are shown in Fig. 8. As seen in Fig. 8, for all pH values, the removal of COD seemed to increase until 0.25 g/L of $Al_2(SO_4)_3$. With higher $Al_2(SO_4)_3$ concentrations, the removal of COD was almost constant.

The coagulation of the clarified effluent was also evaluated using chitosan, a natural polymer obtained from crustacean exoskeletons. The results of COD removal from the clarified effluent with chitosan are presented in Fig. 9. As seen in Fig. 9, chitosan promoted COD removal values from the clarified effluent from 21.5 to 66.6%, depending on the coagulant concentration and adjusted pH. Comparing COD removal efficiency by chitosan and $Al_2(SO_4)_3$, except at pH 7, we found that coagulation with chitosan achieved the best organic matter removal under the tested conditions. The data in Fig. 9 also show that the best result was obtaining using a concentration of 0.80 g/L at pH 9, achieving up to 65% of COD removal. It is worth mentioning that similar results were achieved when treating wastewater from the pulp and paper industry by Thirugnanasambandham et al. (2014). Indeed, these authors claim to have obtained a 66.7% reduction in COD with chitosan (2.0 g/L) at pH 6 and a reduction of 56.9% at pH 8.

It is worth mentioning that after removing 20% of the COD by acidifying the dark liquor and 66.6% by coagulating the clarified effluent with chitosan, the residual COD was still very high (10,015 mg/L). Thus, we decided to perform an oxidizing step by ozonation. For comparison purposes, we also ozonated the dark liquor, the clarified effluent after acidification at pH 3 with HCl and its coagulation supernatant obtained with chitosan (0.80 g/L) at pH 9; the results are presented in Table 9. It is important to highlight that, for the ozonation studies, a new dark liquor was used for each pH level studied. Thus, different values for COD were obtained for the dark liquor, as well as for COD removal after the processes of acidification and coagulation. To better understanding the results of COD removal by ozonation presented in Table 9, the starting COD removal of the dark liquor is given as 0.0%. Note that the values of COD removal at time 0 h correspond to the results obtained by acidifying the dark liquor and coagulating the clarified effluent, prior to ozonation. As shown in Table 9, in almost all cases, an increase in the COD value was observed rather than a decrease. Indeed, with few exceptions (dark liquor at pH 10 and 12), the COD removal presented lower values from the

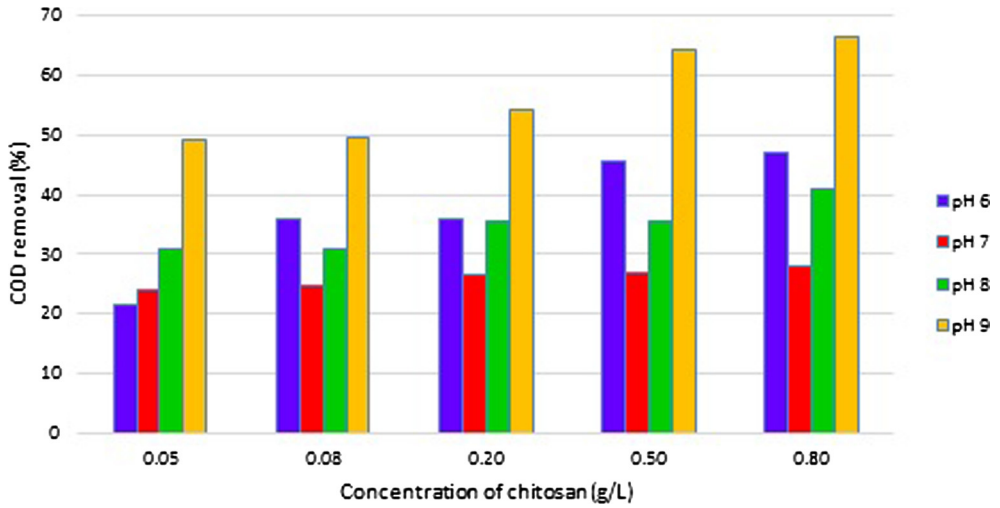


Fig. 9. COD removal during coagulation of the clarified effluent with chitosan.

Table 9
Results of COD removal (%) by ozonation in the dark liquor, the clarified effluent and the coagulation supernatant.

Initial pH	Ozonation time (h)	COD removal (%)		
		Dark liquor	Clarified effluent	Coagulation supernatant
9	0	0.0	20.3	61.1
	1	−10.9	16.8	41.5
	2	−11.1	8.2	38.7
	3	−6.0	9.9	38.6
	4	−8.2	12.1	44.8
10	0	0.0	19.8	62.5
	1	3.3	20.7	39.1
	2	17.3	16.5	42.9
	3	13.5	13.6	38.5
	4	18.9	14.3	43.5
11	0	0.0	19.9	62.3
	1	1.9	10.6	48.4
	2	−1.5	12.7	36.6
	3	2.4	14.4	36.6
	4	0.9	15.1	40.9
12	0	0.0	19.5	59.2
	1	11.3	9.9	45.8
	2	16.9	9.8	44.5
	3	14.5	−4.1	38.2
	4	15.2	5.0	37.9

Table 10
NNK, NNN and nicotine contents in the dark liquor and the different effluents obtained during treatment, along with the literature data for tobacco smoke (Hoffmann et al., 1990).

	NNK	NNN	Nicotine
Tobacco smoke (ng/cig[a]) (Hoffmann et al., 1990)	115.6	133.1	750,000
Dark liquor (ng/mL)	3,057.3	1,635.2	35,400,000
Clarified effluent (ng/mL)	2,014.1	486.5	200,000
Chitosan coagulation supernatant (ng/mL)	11.2	484.3	ND[b]
Ozonation of chitosan coagulation supernatant 1 h (ng/mL)	11.0	483.3	ND
Ozonation of chitosan coagulation supernatant 2 h (ng/mL)	7.4	ND	ND
Ozonation of chitosan coagulation supernatant 3 h (ng/mL)	5.2	ND	ND
Ozonation of chitosan coagulation supernatant 4 h (ng/mL)	0.8	ND	ND

[a] cig: cigarette.
[b] ND: not detected.

starting levels. This behavior has been observed by other authors and is probably the result of incomplete mineralization of the organic material in the effluent, generating recalcitrant intermediates, as found by Melo et al. (2009). Thus, ozonation, under the conditions studied here, showed no efficiency in the treatment of the effluent.

The different effluents were also analyzed in terms of the metal contents and nitrogenated compounds. As shown in Table 7, the dark liquor presented very low contents of Cr, As, Cd, Pb, Ni and Se, below the limits established by Brazilian law. After acidification to precipitate the lignin, the clarified effluent presented undetectable metal contents, even for As and Cr. Table 10 presents the results for the most toxic nitrogenated compounds (NNK, NNN and nicotine) in the dark liquor, clarified effluent, supernatant from coagulation with chitosan and after ozonation at pH 12, compared with content of these compounds in cigarette smoke (Hoffmann et al., 1990). As can be noted in Table 10, the content of the three nitrogenated compounds studied here strongly decreased with the treatments. NNK was the only compound still detected after ozonation, although at a very low level. NNN was undetectable after 2 h of ozonation and nicotine was not quantified in the effluent after coagulation. These results indicate that the treatments were effective for the removal of these compounds, which can be confirmed by the reduction in organic matter expressed as COD after each treatment.

3.3. Cellulose pulp production from processed cigarette butts

Since we could not efficiently reduce the COD in cigarette butt cooking effluent, we aimed to develop an environmental friendly process to reuse the clarified effluent after acidifying the dark liquor to pH 3. Therefore, we propose that, after decantation of the precipitate, the clarified effluent should be collected and then, after adjusting its pH to 7, used in a new pulping process, as shown in Fig. 10.

We repeated the process seven times. Fig. 11 shows the dark liquor samples and the clarified effluent in each cycle. As can be seen in Fig. 11, the appearance of the samples remained almost constant after all cycles.

Table 11 presents the COD values for each cycle of effluent reuse. As shown in Table 11, no change in the pH of dark liquor was observed, since the clarified effluent had the same pH corrected by the additional water used for the pulping process. Regarding COD, an increase was observed for the initial reuses, but we noted that the COD value stabilized from the fifth reuse on.

The process developed here was licensed to a recycling company (Poiato Recicla, Sao Paulo, Brazil), which has started up a pilot plant in March, 2016. It is worth mentioning that they have operated the process several times, always reusing the clarified effluent, and have already processed some tons of cigarette butts. Indeed, they are obtaining a good quality cellulose pulp which is sold to a cardboard factory.

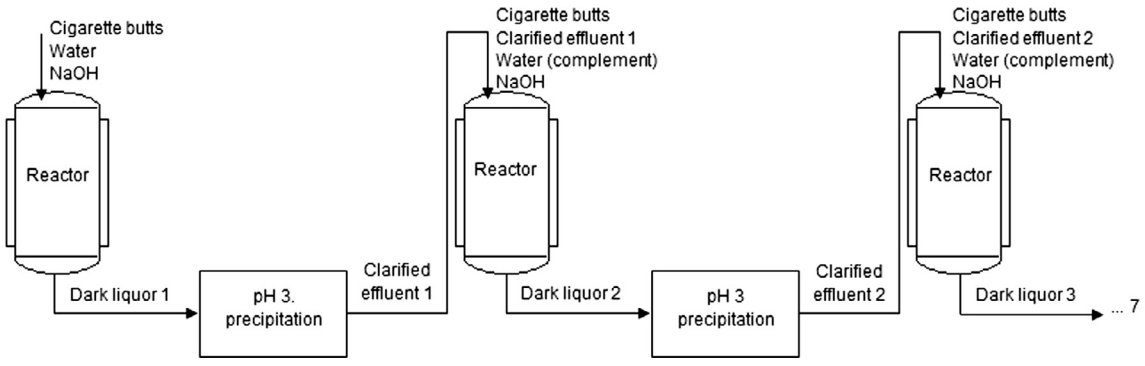


Fig. 10. Proposed flowchart for cellulose pulp production from cigarette butts by reusing the effluent from the precipitation process at pH 3.

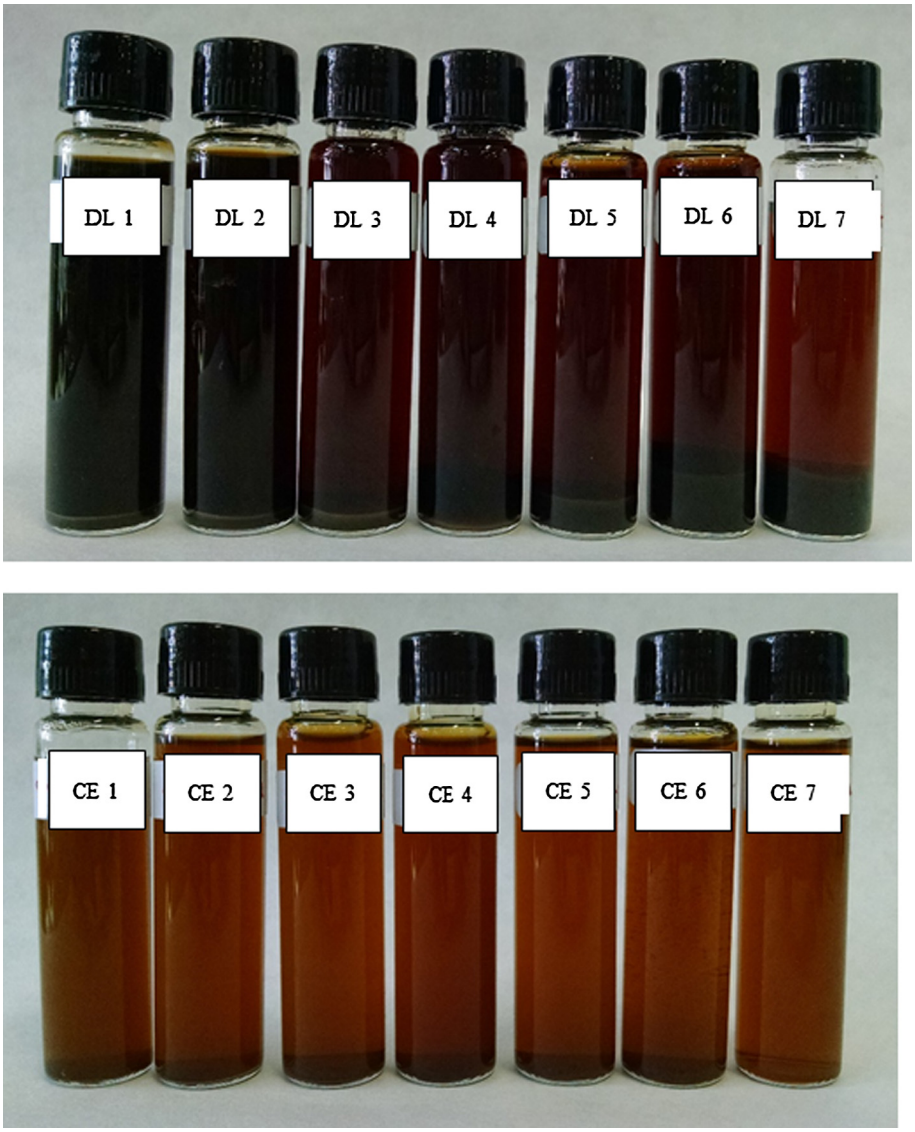


Fig. 11. Appearance the dark liquor (DL) and clarified effluent (CE) over seven cigarette butt cooking cycles reusing the effluent.

Table 11
Dark liquor and clarified effluent pH and COD values after seven cycles of reusing the effluent.

Sample	pH	COD (mg/L)
Dark liquor 1	9.1	31,943.3 ± 72.3
Clarified effluent 1	3.0	26,023.3 ± 47.3
Dark liquor 2	8.9	39,940.0 ± 40.0
Clarified effluent 2	3.0	37,680.0 ± 87.2
Dark liquor 3	9.0	43,486.7 ± 70.2
Clarified effluent 3	3.0	43,300.0 ± 91.7
Dark liquor 4	8.5	52,233.3 ± 75.7
Clarified effluent 4	3.0	52,033.3 ± 30.6
Dark liquor 5	8.8	60,220.0 ± 87.2
Clarified effluent 5	3.0	60,046.7 ± 41.6
Dark liquor 6	9.2	60,713.3 ± 50.3
Clarified effluent 6	3.0	60,660.0 ± 34.6
Dark liquor 7	8.7	60,653.3 ± 30.6
Clarified effluent 7	3.0	60,700.0 ± 52.9

4. Conclusions

In summary, we have shown that is possible to produce cellulose pulp from cigarette butts. The produced cellulose pulp may be absorbed by the recycling paper industry to be used pure or blended with pulps from various origins according to the quality requirements of the final paper. Thus, the process proposed here consists of an environmentally friendly solution for this solid residue which is generated in large quantities throughout the world. The dark liquor produced as the effluent in the pulping process contains a very high resistant organic content, including toxic nitrogenated compounds. Even after three treatment processes (precipitation by acidification, coagulation and ozonation), it was not possible to reduce the COD value of the effluent to an acceptable value. However, it was possible to reuse the effluent after one treatment step (precipitation) in the pulping process, thereby eliminating any liquid effluent from the overall process. Finally, it is worth mentioning that the process developed here was licensed to a recycling company, which has recently started up a pilot plant.

Acknowledgments

The authors are in debt to CAPES, CNPq and FAPDF for partial financial support. PAZS thanks CNPq for the research fellowship.

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